GEOG 4/5/7 9073: Environmental Analysis in R

Week 05.01: R as a GIS

Dr. Bitterman

Today's schedule

- Open discussion
- Some basic spatial operations
- Activity

Anything to discuss? Questions?

This week's focus:

getting comfortable with some simple geospatial operations

today's setup

- there's new data in the `./data/` directory of the repository, and we'll be using some new packages as well

```
library(tidyverse)
library(sf)
library(GISTools)
library(tmap)
```

Let's start with some simple data

Break it down first

```r
# let's start with some data

streams <- sf::read_sf("./data/Streams_303_d_.shp")
tm_shape(streams) + tm_lines()
```

what happened?

More data...

```r
counties <- sf::read_sf("./data/County_Boundaries-_Census.shp")

counties_areas <- sf::st_area(counties)
```

and what happened here?

If you wanted to, how would you add the areas back to the sf data.frame?

Let's review... how would you use your data and functions to "get" ONLY Lancaster County?

Something like this:

which function are we using, and how does it work?

```
lc <- counties %>% dplyr::filter(., NAME10 == "Lancaster")
```

How would we find only those 303d streams in Lancaster County?

Something like this:

```
lc_303ds <- sf::st_intersection(streams, lc)
```

Plot them:

Here's some options. Let's break them down - how are they different?

```
tm_shape(lc_303ds) + tm_lines()
tm_shape(lc_303ds) + tm_lines(col = "blue")

tm_shape(lc_303ds) + tm_lines(col = "Waterbody_")
```

What happened?

Buffers: a simple spatial operation

- What's a buffer?

Let's try one

Break down the code...

```
buffs <- sf::st_buffer(lc_303ds, dist = 1000)

tm_shape(buffs) + tm_polygons(col = "Waterbody_")
```

What does it look like?

How MUCH did we buffer the streams?

Fill vs. color

For areal units...

```
tm_shape(buffs) + tm_polygons(col = "Waterbody_")

tm_shape(buffs) + tm_polygons(fill = "Waterbody_")
```

What's the difference?

Let's try some point data

```r
# first read the state parks shapefile
parks <- sf::read_sf("./data/State_Park_Locations.shp")

# subset to lancaster county
lc_parks <- sf::st_intersection(parks, lc)

# plot them
tm_shape(lc_parks) + tm_dots(col = "AreaName", size = 1)
```

Plotting multiple layers using `tm`

```
tm_shape(lc_303ds) + tm_lines(col = "Waterbody_") +
  tm_shape(lc_parks) + tm_dots(fill = "AreaName", size = 1)
```

What happened?

Your final task

- find all of the state parks within 0.5 miles of a 303d stream
- plot just those points WITH the corresponding stream segment
- use color to distinguish the points and stream segments

GO!

For this week

- Chapters 5 & 6
- Practice, practice, practice
- Lab 01 due